

Topics:

- + System.
- + Body language and proxemics.*****
- + Three aspects of body language.
- + Spatial zones.*****
- + System requirements.
- + Types of requirements.*****
- + Type of non-functional requirements.
- + Criteria to define system requirements.
- + The process of requirements discovery.
- + Seven fact-finding methods(Questionnaires,interviews).
- + Questionnaires.*****
- + Interviews.*****
- + Interviewing Do's and Don'ts.*****
- + Communicating.
- + Brainstorming.*****
- ❖ **System:** Means an organized relationship among functioning units or components.
- ❖ **Body language and proxemics:** Form of nonverbal communication that
 - ✓ We all use & are usually unaware of
 - ✓ By research
 - Verbally-7 % (in word)
 - Tone of voice- 38 %
 - Facial & Body expressions- 55%
 - **If you hear only word you will miss what you want to say.**
- ❖ **Three aspects of body language.**
 - **Facial disclosure**
 - One of the most controlled parts of body
 - **Eye contact**
 - Direct eye contact can cause strong feelings, either positive or negative
 - **Posture (attitude)**
 - Least controlled aspect of the body
- ✓ A good analyst will watch the audience for changes In posture that could Indicate anxiety, disagreement, or boredom.

❖ **Spatial zones**

- ✓ Intimate zone—closer than 1.5 feet
- ✓ Personal zone—from 1.5 feet to 4 feet
- ✓ Social zone—from 4 feet to 12 feet
- ✓ Public zone—beyond 12 feet

❖ **System requirements:** A requirement may describe functions, features (attributes), and constraints.

❖ **Types of requirements.**

1. **Functional requirements:** A functional requirement is a function or feature that must be included in an information system in order to satisfy the business need and be acceptable to the users.
2. **Non-functional requirements:** A nonfunctional requirement is a description of the features, characteristics, and attributes of the system as well as any constraints that may limit the boundaries of the proposed solution.

❖ **Types of non-functional requirements:**

Economy	Economy requirements represent the need for the system to reduce costs or increase profits. • What are the areas of the system where costs must be reduced? • How much should costs be reduced or profits be increased? • What are the budgetary limits? • What is the timetable for development?
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❖ **Criteria to define system requirements:**

- ✓ Consistent (reliable)
- ✓ Complete
- ✓ Feasible
- ✓ Required
- ✓ Accurate
- ✓ Traceable
- ✓ Verifiable

❖ **The process of requirements discovery.**

- ✓ Problem discovery and analysis.
- ✓ Requirements discovery.
- ✓ Documenting and analyzing requirements.
- ✓ Requirements management.

❖ Seven fact-finding methods(Questionnaires, interviews).

- ✓ Sampling of existing documentation, forms, and databases.
- ✓ Research and site visits.
- ✓ Observation of the work environment.
- ✓ **Questionnaires.**
- ✓ **Interviews.**
- ✓ Prototyping.
- ✓ Joint requirements planning (JRP).

❖ **Questionnaires.**

Questionnaires are special-purpose documents that allow the analyst to collect information and opinions from respondents.

➤ **Advantages of questionnaires:**

- Can be answered quickly.
- Inexpensive means.
- Real facts expressed.
- Responses can be tabulated and analyzed quickly.

➤ **Disadvantages of questionnaires:**

- Respondents often low
- No guarantee to get answer or expand all question
- No opportunity for voluntary information
- Not possible to read body language
- Good questions are difficult to prepare

➤ **Types of fixed-format questions:**

- ✓ Multiple-choice questions
- ✓ Rating questions
- ✓ Ranking questions

➤ **Questionnaire procedure:**

- I. Determine what facts and opinions must be collected and from whom you should get them.
- II. Based on the needed facts and opinions, determine whether free- or fixed-format questions will produce the best answers.
- III. Write the questions.
- IV. Test the questions on a small sample of respondents.
- V. Duplicate and distribute the questionnaire.

❖ **Interviews:** Interviews are a fact-finding technique whereby the systems analysts collect information from individuals through face-to-face interaction.

➤ **Advantages of interviews:**

- ✓ Opportunity to motivate the interviewee
- ✓ More feedback from the interviewee
- ✓ Permit system analyst to adapt or reword questions for each individual
- ✓ possible to read body language

➤ **Disadvantages of interviews:**

- ✓ Time consuming
- ✓ Success highly depends on system analysts' human relation skill
- ✓ May be impractical due to location

➤ **Types of interview questions:**

- **Open-ended** questions allow the interviewee to respond in any way that seems appropriate.
- **Closed-ended** questions restrict answers to either specific choices or short, direct responses.

➤ **Procedure and interview questions:**

- ✓ Select Interviewees
- ✓ Prepare for the Interview
 - An interview guide is a checklist of specific questions the interviewer will ask the interviewee.
- ✓ Conduct the Interview
- ✓ Follow Up on the Interview

➤ **Types of Questions to Avoid**

- ✓ Loaded questions.
- ✓ Leading questions.
- ✓ Biased questions.

➤ **Interview Question Guidelines**

- ✓ Use clear and concise(short) language.
- ✓ Don't include your opinion as part of the question.
- ✓ Avoid long or complex questions.
- ✓ Avoid threatening questions.
- ✓ Don't use "you" when you mean a group of people.

❖ Interviewing Do's and Don'ts:

Do	Avoid
• Be courteous (polite) • Listen carefully • Maintain control • Probe (inquiry) • Observe mannerisms and nonverbal communication • Be patient • Keep interviewee at ease • Maintain self-control	• Continuing an interview unnecessarily. • Assuming an answer is finished or leading nowhere. • Revealing (express) verbal and nonverbal clues. • Revealing your personal biases. • Talking instead of listening. • Assuming anything about the topic and the interviewee. • Tape recording -- a sign of poor listening skills.

❖ **Listening:** "To hear is to recognize that someone is speaking, to listen is to understand what the speaker wants to communicate."

❖ Guidelines for communicating:

- Approach the Session with a Positive Attitude
- Set the Other Person at Ease
- Let Them Know You Are Listening
- Ask Questions
- Don't Assume Anything
- Take Notes

❖ **Brainstorming:** Brainstorming is a technique for generating ideas during group meetings.

❖ Brainstorming guidelines:

- ✓ Isolate the appropriate people in a place that will be free from distractions (diversion) and interruptions
- ✓ Make sure that everyone understands the purpose of the meeting
- ✓ Appoint one person to record ideas
- ✓ Remind everyone of the brainstorming rules
- ✓ Within a specified time period, team members call out their ideas as quickly as they can think of them
- ✓ Refine, combine, and improve the ideas that were generated earlier